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Electrolyte loaded hexagonal boron nitride/polyacrylonitrile nanofibers for lithium ion battery application

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ABSTRACT

In the present work, Novel hBN/polyacrylonitrile composite nanofibers were produced via electrospinning approach and loaded with electrolyte for rechargeable lithium-ion battery applications. The electrospun nanofibers comprising various hBN contents were characterized by using Fourier transform infrared spectroscopy (FT-IR), thermogravimetric analysis (TGA), X-ray diffraction (XRD) and scanning electron microscopy (SEM) techniques. The influence of hBN/PAN ratios onto the properties of the porous composite system, such as fiber diameter, porosity and the liquid electrolyte uptake capability were systematically studied. Ionic conductivities and electrochemical characterizations were evaluated after loading electrospun hBN/PAN composite nanofiber with liquid electrolyte, i.e., 1 M lithium hexafluorophosphate ($LiPF_6$) in ethylene carbonate (EC)/ethyl methyl carbonate (EMC) (1:1 vol). The electrolyte loaded nanofiber has a highest ionic conductivity of 10^{-3} S cm^{-1} at room temperature. According to cyclic voltammetry (CV) results it exhibited a high electrochemical stability window up to 4.7 V versus Li^+/Li. Li//10 wt% hBN/PAN//$LiCO_2$ cell was produced which delivered high discharge capacity of 144 mAhg^{-1} and capacity retention of 92.4%. Considering high safety and low cost properties of the resulting hBN/PAN fiber electrolytes, these materials can be suggested as potential separator materials for lithium ion batteries.

1. Introduction

Li-ion batteries have become important in the field of electronic industry due to their advantages like compactness, lightweight, high operational voltage and providing highest energy density [1–3]. Typical Li-ion battery has a cathode ($LiCoO_2$, $LiMnO_2$, $LiFePO_4$ etc.), an anode (graphite, graphene, carbon nanotubes, carbon nanofibres, lithium titanium oxides etc.) and a separator [1,3].

The separator provides an electrical insulation between anode and cathode and allow ion transfer during operation. It also plays a significant role in determining battery performance [4,5]. The performance of the Li-ion battery separator is determined by several factors such as permeability, porosity, electrolyte uptake capacity, mechanical, thermal and chemical stability [1,3,6]. Several commercially available polymers have been used as separators and the most common polymers are poly(ethylene) [7], poly(propylene) [8], poly(ethylene oxide) [9], poly(acrylonitrile) [10–12], poly(methyl methacrylate) [13] and poly (vinylidene fluoride) (PVDF) [14–16]. During the preparation of the battery separator several types of Li-salts ($LiClO_4$, $LiPF_6$, CF_3SO_3Li etc.) and fillers (SiO_2, TiO_2, etc.) [9,11,17,18], molecular sieves and clays [19,20], ferroelectric materials [21] and carbon based fillers can be mixed with the polymer matrix. Although Li-salts participate in the conduction process, fillers are not directly contributing to the conduction process. They enhance of morphological, mechanical, thermal, electrochemical properties and interfacial stability of the separators [4,5].

Hexagonal boron nitride (hBN) is physically analogous to graphite with comprising hexagonal layers of 0.33 nm distance and is the most examined polymorph among other systems [22]. It is white powder and has a melting point of 3000 °C. hBN has important physical and chemical properties such as high corrosion resistance, good lubrication property, high thermal conductivity, high temperature stability, good resistance to oxidation and chemical inertness. It has several applications such as coatings, electrical insulation, optical storage, optoelectronic devices, medical treatment, and lubricant [23–26]. The blending of hBN in polymers is mostly due to its thermal conductivity property [24]. The usage of hBN in fuel cell polymer electrolyte was previously investigated by our group and contribution of hBN to the electrolyte stability as well as proton conduction was reported [27,28]. Another important innovative part of this work is that the presence of high

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thermal conducting separator may positively contribute to the thermal management in lithium ion batteries.

Polyacrylonitrile (PAN) is a semicrystalline thermoplastic polymer used in many industrial areas. It provides several improvements with easy modification of its morphology and physical properties [10,11,17].

Electrospinning of polymer solutions to prepare thin fibers provides high porosity, better morphology and high surface area resulting high electrolyte uptake and easy ion transport [14]. If the processing parameters are controlled it is possible to obtain electrospun fibrous materials with 30–90% porosity and μm range pore sizes [15]. It is preferred in recent years for the production of polymer nanocomposites due to the elimination of homogeneity problems observed in normal casting methods.

The present work includes the preparation of new hBN/PAN composite nanofibers by electrospinning. The nanofibers were characterized with different spectroscopic techniques. The morphological properties and the electrochemical performance of the liquid electrolyte loaded hBN/PAN composite nanofibers were investigated.

The Li ion conductivity of electrolyte loaded nanofibers were measured via dielectric-impedance analyzer. A battery cell, Li//10 wt% hBN/PAN//$LiCO_2$ was constructed and its capacity as well as cycling stability results are discussed.

2. Experimental

2.1. Materials

PAN (average Mw = 150.000 g mol^{-1}) and *N*,*N*-Dimethylformamide(DMF) were obtained from Sigma-Aldrich. Liquid electrolyte, 1 M lithium hexafluorophosphate ($LiPF_6$) in ethylene carbonate and ethyl methyl carbonate (EC + DEC, 1:1 by volume), was delivered from Sigma Aldrich. Hexagonal boron nitride (hBN, particle size = ~70 nm) was supplied from Lower Friction. $LiCoO_2$, Lithium chip and CR2032 cells were purchased from MTI Corporation. All chemicals were used without further purification. Lithium chip and electrolyte solution were stored in the Argon filled glove box.

2.2. Preparation of electrospun hBN/PAN nanofibers

Polyacrylonitrile (PAN) and Hexagonal boron nitride (hBN) was vacuum dried at 60 °C for 8 h before use. Polyacrylonitrile was used as a matrix polymer. Hexagonal boron nitride was used as a filler in preparing composite nanofibers. *N*,*N*-dimethylformamide was used as the solvent of the polymer solutions. Composite materials were prepared by electrospinning at room temperature. In order to obtain nanofiber, PAN solution (8%) was prepared by dissolving the polymer in DMF. hBN/PAN solutions were prepared by adding different amounts of hBN (3, 5, 7 and 10 wt%) into the 8 wt% PAN solution. Prior to electrospinning all the solutions were mechanically stirred overnight.

Electrospinning of the homogeneous solution was carried out at a flow rate of 2 ml h^{-1} with a high voltage of 20 kV at room temperature. A space of 20 cm was arranged between the syringe tip and collector sheet. The electrospun fibers were collected on aluminum foil, and were dried in vacuum at 60 °C for 12 h before further use.

To obtain electrolyte loaded nanofiber, dried porous fibrous materials were dipped into liquid electrolyte (1 M $LiPF_6$ in EC/DMC (1:1, v,v)) at room temperature in a glove box for 1 h. Then the surface was wiped off with filter papers to get rid of excess electrolyte on the surface.

2.3. Material characterization

Prior to all measurements, the electrospun nanofibers and electrolyte loaded nanofibers were dried at 60 °C under vacuum and stored in a glove box. They were characterized for thickness, surface morphology, spectral properties and porosity.

The IR spectra (4000–400 cm^{-1}, resolution 4 cm^{-1}) were recorded with a Bruker Alpha- P in ATR system.

Thermal stabilities of the polymer electrolytes were examined by thermogravimetry analysis (TGA) with a Perkin Elmer STA 6000. The samples (~3 mg) were heated from room temperature to 750 °C under N_2 atmosphere at a heating rate of 10 °C·min^{-1}.

The surface morphologies of polymer electrolytes were observed by scanning electron microscopy (SEM, Philips XL30S-FEG). All of the samples were sputtered with gold for 150 s before SEM measurements.

X-ray data were obtained from fully automated 9.0 kW anode rotating generator Rigaku Smart Lab X-Ray X-Ray Diffractometer. Θ angle is between 0 and 90°.

The following equation is used to calculate the amount of liquid electrolyte uptake:

$$\text{Electrolyte Uptake } (\%) = \frac{W_w - w_d}{w_d} \times 100$$

where w_d and w_w are weights prior and after absorbing the electrolyte solution, respectively. Weights were measured in a glove box.

The porosities of the hBN/PAN nanofiber systems were determined by using n-butanol uptake tests. After the uptake test, the following equation was used to calculate the porosity of the materials.

$$\text{Porosity } (\%) = \frac{w_w - w_d}{\rho_b \times V} \times 100$$

where w_w and w_d are the weights of wet and dry materials, respectively, ρ_b the density of n-butanol, and V the geometric volume of the material.

The alternating current (AC) conductivities of the electrolyte loaded samples were determined using a Novocontrol dielectric-impedance analyzer within the frequency range from 0.1 Hz to 3 MHz with respect to temperature. The films with a diameter of 10 mm and a certain thickness were placed between platinum blocking electrodes and their conductivities were measured at 10 K intervals under a dry nitrogen atmosphere.

Electrochemical stability was measured by a linear sweep voltammetry (LSV) of a Li/PE/SS cell using Metrohm Autolab (Multichannel-M101 Model) electrochemical analyzer at a scan rate of 1 mV s^{-1}, with voltage from 2.0 to 6 V.

The Li/PE/$LiCoO_2$ cell systems were assembled for charge and discharge cycling meaurements. An automatic charge-discharge unit, MTI battery analyzer system (model BST8-MA) was used for electrochemical performance. The tests were done between 2.5 and 4.2 V at room temperature and at a current density of 0.1 C.

3. Results and discussions

3.1. FT-IR spectra

The FI-IR spectra of pure PAN fibers and hBN/PAN nanocomposite fibers are illustrated in Fig. 1. At around 2935 cm^{-1} a peak is observed which is due to the methylene ($-CH_2-$) group, the peaks at 2246 cm^{-1} and 1449 cm^{-1} can be attributed to stretching vibration of nitrile groups ($-CN-$) and the bending vibration of methylene ($-CH_2-$), respectively [29]. The FT-IR spectra of all the samples after spinning are showing the presence of characteristic peak of $-C{\equiv}N$ at 2243 cm^{-1}.

Huang et al. previously studied the FT-IR spectra of hBN [28,30]. According to Huang et al. the hBN nanosheets have primary and secondary amine groups and also hydroxyl groups at the edge, but these groups are very less hence can't be detected in the presence of strong B—N bands. hBN has two intense bands which belong to B—N out of plane bending vibration and B—N bending at 1336 and 760 cm^{-1}, respectively [31,32].

The nanofibers exhibit characteristic hBN peaks at 798 and 1360 cm^{-1}. All the electrospun samples show characteristic peak of $-C{\equiv}N$ at 2243 cm^{-1}. From the obtained results it can be concluded that the applied electrical field didn't cause change in the

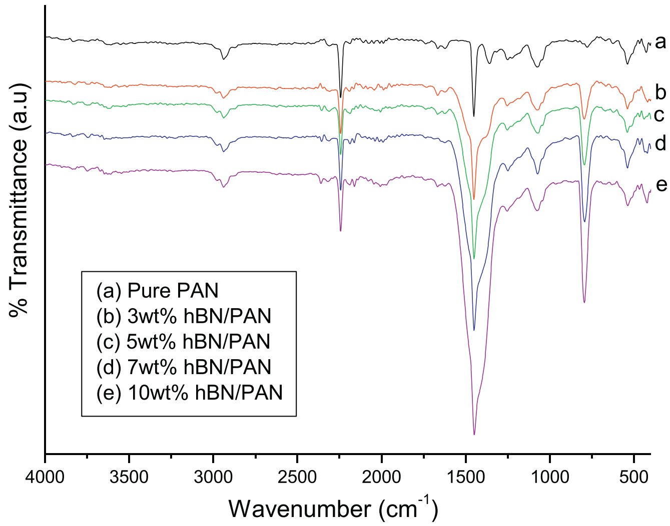

Fig. 1. FT-IR spectra of electrospun hBN/PAN composite materials with various hBN contents.

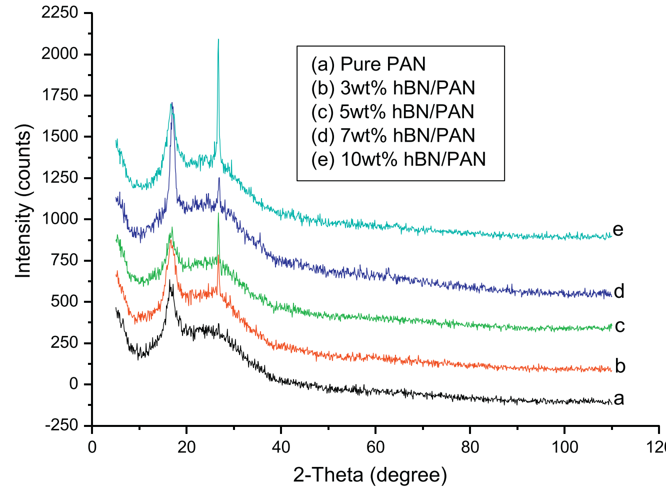

Fig. 3. XRD patterns of electrospun hBN/PAN composite materials with various hBN contents.

electrospinning process and the composite nanofibers are successfully obtained [10].

3.2. Thermal properties

Thermal stability happens to be one of the important property for a polymer electrolyte during application in lithium ion batteries [16,33]. By employing TGA the thermal stability of the electrospun nanofibers are deduced under N_2 atmosphere from room temperature to 750 °C at a heating rate of 10 °C min^{-1}. Fig.2 illustrates the obtained results. From Fig. 2 it is very clear that the electrospun nanofibers are stable up to 280 ± 5 °C, all the samples exhibit similar thermal stability. It is also seen that as the amount of hBN increases the first weight loss temperature increases as well. Results indicate hBN has a significant effect on the thermal properties of the composite films. The char starts yielding at 750 °C. As the amount of hBN increases in the composites the more char residue formed [24].

3.3. XRD pattern of hBN/PAN composites

Fig. 3 shows the XRD pattern of pure PAN and hBN/PAN nonofiber

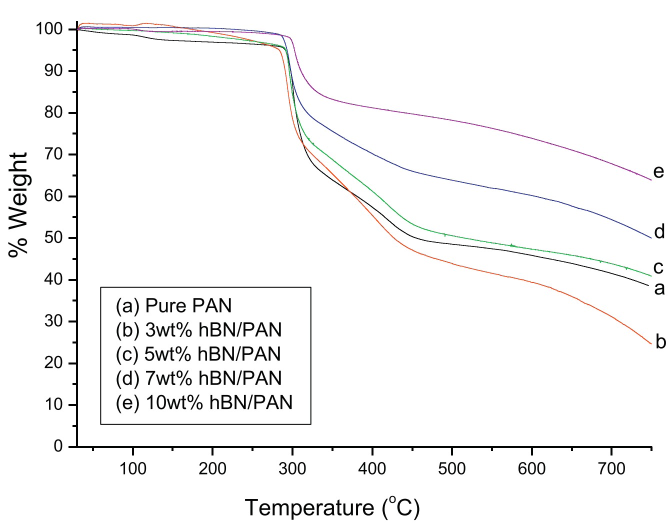

Fig. 2. TG thermograms of electrospun hBN/PAN composite materials with various hBN contents at a heating rate of 10 °C min^{-1} in an inert atmosphere.

composites with various hBN contents. The XRD pattern of electrospun pure PAN exhibits a sharp peak at 17° which is related to the (010) plane with a *d*-spacing of 5.30 °C, demonstrating the crystalline behavior of PAN [34]. Pure PAN was reported to have another peak at 28° [34] however this peak is relatively weak and broad in our sample which may be due to the enhanced amorphous region in PAN nanofibers [11]. hBN was reported to have a strong peak at 27° and a small peak at 42° [35]. As seen in Fig. 3 there is a strong peak at 27° in composites due to the presence of hBN. There is no decrease in the intensity of PAN peak at 17° which indicates that the electrospun composite have crystalline morphology. It is reported that small inorganic fillers may decrease the crystalline behavior of PAN polymer [11,17] but in that study hBN does not decrease the crystalline nature of the PAN fiber which may be due to the high particle size (70 nm) of hBN.

3.4. Nanofiber morphology

In the electrospinning technique, the inclusion of hBN particles has significant role on the morphology of the fiber such as surface roughness, fiber diameter and orientation [11]. Fig. 4 illustrates SEM images and fiber diameter distributions of pure PAN and hBN/PAN composite fibers with various hBN contents. Pristine PAN fiber exhibits relatively uniform and smooth fibers. Its average fiber diameter is about 220 nm with a distribution of 125–350 nm. Besides, fibers in hBN/PAN composite system represent irregularity and defects on the fiber surface. The average fiber diameter (Table 1) increases from 220 to 385 nm for 3–7 wt% samples and decreases to 250 nm for 10 wt% sample. However, there are several nodal points for that sample which may be due to the aggregation of the hBN particles.

Table 1 shows the porosity and average fiber diameter of hBN/PAN composite fibers. The average fiber diameter has a considerable relation with the material porosity, pore size, and specific surface area. The electrolyte solution uptake, electrochemical performance and stability are also related with the mentioned factors. The porosity, measured by n-butyl alcohol (BuOH) uptake, decreases from 241% (PAN) to 115% (PAN7hBN) while average fiber diameter increasing from 220 (PAN) to 385 nm (PAN7hBN). Although the fiber structure is not homogeneous, the porosity increases to 204% and the fiber diameter decreases to 250 nm for 10 wt% hBN. The samples exhibit much higher porosity than the reported electrospun PAN fibers [11,34,36]. As the fiber diameter decreases porosity increases due to the presence of more void space between thin fibers [39]. Hence, the introduction of hBN nanoparticles decreases the porosity value by increasing the fiber diameter.

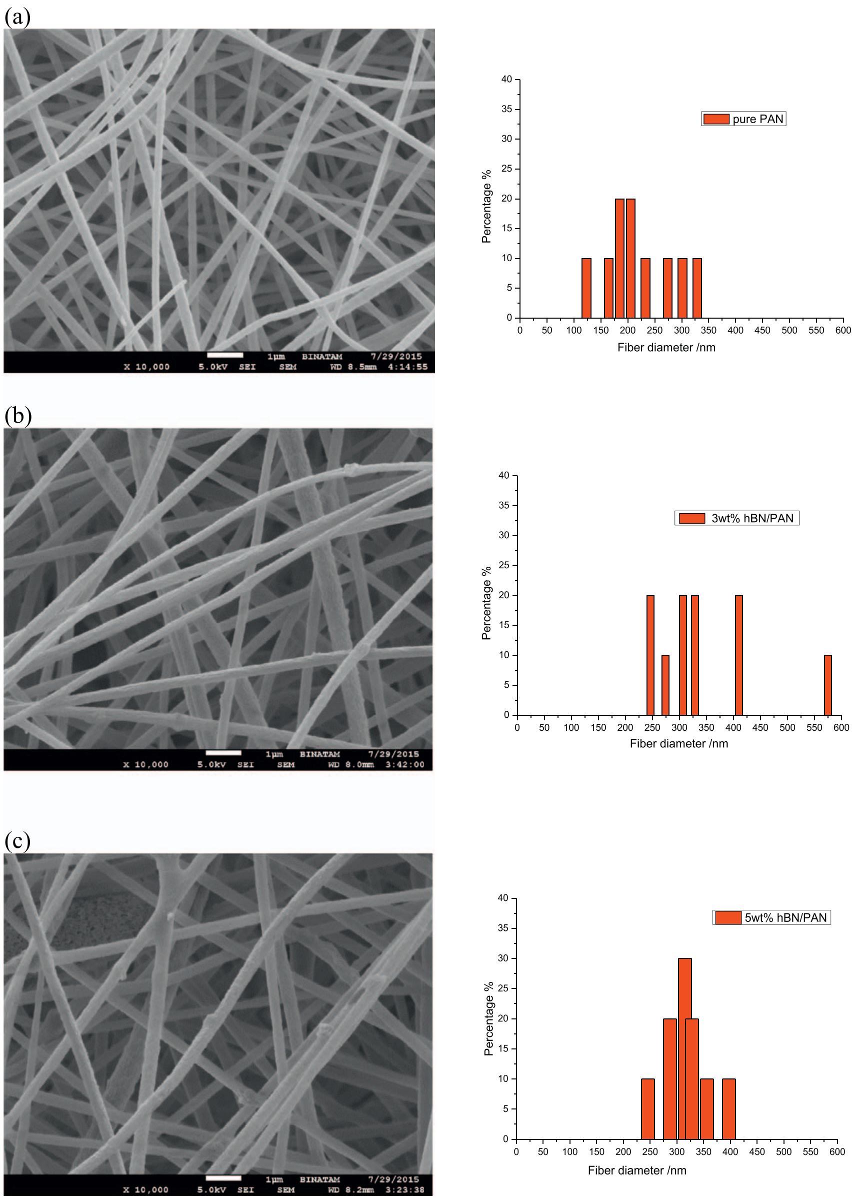

Fig. 4. SEM images and fiber diameter distribution of electrospun hBN/PAN composites with various hBN contents: (a) pure PAN, (b) 3 wt%, (c) 5 wt%, (d) 7 wt%, and (e) 10 wt%.

3.5. Electrolyte uptake for electrolyte loaded electrospun hBN/PAN composite

The rapid and high amount of electrolyte uptake capacity of separators is important for battery application since it can diminish the electrolyte filling time and improve the cell performance [3,36,37]. Electrolyte uptake capacities of hBN/PAN systems with different hBN content are shown in Table 1. The uptake capacities of 0, 3, 5, 7, and 10 wt% hBN/PAN are 1390, 1560, 1790, 1315 and 1250%, respectively. The electrolyte uptake of materials initially increases with lower hBN content (3, 5%) and then partially decreases with higher hBN content (7, 10%). This uptake capacity can be said to be sufficient for separator usage.

Table 1
Average fiber diameter, porosity and electrolyte uptake properties of the fibers.

Sample	Avr. fiber diameter (nm)	Porosity %	Electrolyte uptake (%)
PAN	220	241	1390
3 wt% hBN/PAN	343	206	1560
5 wt% hBN/PAN	318	147	1790
7 wt% hBN/PAN	385	115	1315
10 wt% hBN/PAN	250	204	1250

3.6. Ionic conductivity for electrolyte loaded electrospun hBN/PAN composites

Parameter such as porosity, tortuosity, and thickness of the separator and the resistivity of the electrolyte all affect the ionic conductivity [6,36]. The temperature dependencies of the ionic conductivity of the liquid electrolyte soaked hBN/PAN fiber at room temperature were illustrated in Fig. 5. Ionic conductivity strongly depends on the number of charge carriers and their mobility. As the temperature increases the mobility increases as well this leads to increased ionic conductivity [20]. At room temperature 10 wt% hBN/PAN system has an ionic conductivity of $1.0 \times 10^{-3}\,\mathrm{Scm}^{-1}$. The composite materials have better ionic conductivity compared to literature. A previous study showed that hBN does not block the ionic conduction but enhances slightly [28]. It is also know that inorganic

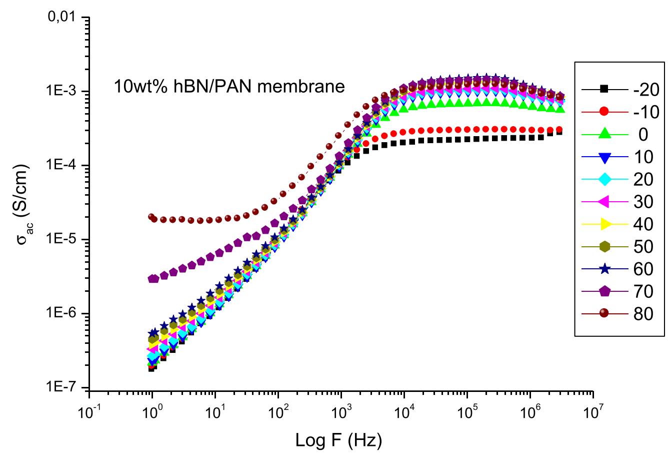

Fig. 5. AC conductivity versus frequency of electrolyte loaded electrospun 10 wt% hBN/PAN composite at various temperatures.

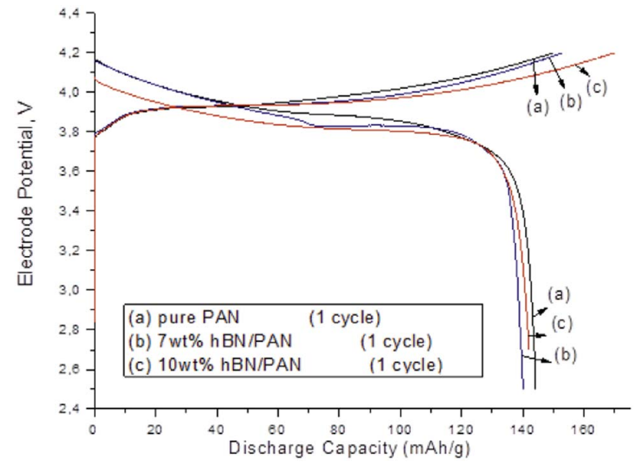

Fig. 7. First-cycle charge-discharge curves of Li/$LiCoO_2$ cells containing electrolyte loaded hBN/PAN nanofiber with different hBN contents at 0.1C.

particles can interact with the electrolyte components, while improving the ionic conductivity [37]. Similar improvement in ionic conductivities was also reported for electrospun SiO_2/PAN systems [11].

3.7. Electrochemical stability for electrolyte loaded electrospun hBN/PAN composite

Linear sweep voltammetry analysis is a very useful method to identify the electrochemical oxidation limits of separators soaked in liquid electrolyte. Fig. 6 shows the voltammetry curves of electrolyte loaded pure PAN and hBN/PAN composite nanofibers. On the linear sweep voltammetry curves, electrochemical oxidation limit of the nanofiber can be determined by the fast increase in the current, which shows beginning of electrolyte decomposition. From Fig. 6 it is observed that the electrochemical oxidation limit of 10 wt% hBN/PAN nanofiber (4.7 V) is higher than that of pure PAN nanofiber (4.3 V). Improvement in the electrochemical oxidation limit could be due to the excellent affinity of hBN nanoparticles to the electrolyte and the stabilization effect of hBN nanoparticles. It has been already reported that SiO_2 nanoparticle embedded nanofiber separators can absorb impurities in the electrolyte and hence can help to stabilize the liquid electrolyte at higher voltages [11,37,38]. The hBN nanoparticles in our composite nanofiber structure might show the similar effect of the SiO_2

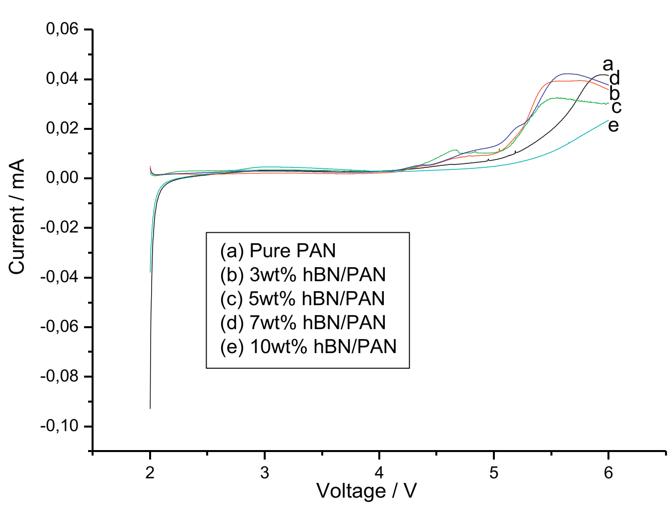

Fig. 6. Electrochemical stability by linear sweep voltammetry of electrolyte loaded electrospun hBN/PAN fiber with different hBN content. (Li/GPE/SS cells, 1 mVs^{-1}, 2–6 V).

nanoparticles. Considering that the practical working potential range of commercial Li-ion batteries is typically between 1.8 and 3.5 V, introduced hBN/PAN composite nanofiber have enough electrochemical stabilities for Li-ion battery application.

3.8. Electrochemical performance of electrolyte loaded electrospun hBN/PAN composite in lithium-ion cell

In order to investigate the electrochemical performance of hBN/PAN composite nanofiber for use as separators in rechargeable Li-ion batteries, coin-type half cells were constructed with $LiCoO_2$ as the cathode and lithium metal as the counter electrode. For comparison, coin-type half cells using PAN nanofiber separator were also fabricated. Fig. 7 shows the first-cycle charge and discharge capacities of PAN and hBN/PAN composite nanofiber at room temperature. Although the capacity performance of the batteries mainly depends on the type and structure of the electrode materials, the structure of separators has a critical impact in determining the overall battery performance because separators can affection transportation efficiency between the two electrodes, which is crucially important in regulating the cell kinetics [38]. It is seen that the initial specific discharge capacities of all cells are between 140 and 144 mAh g^{-1}. From Fig. 7, it is also seen that the first-cycle discharge capacity increases when the hBN nanoparticles are introduced into the nanofibers. Among all the electrolytes studied, the cell containing 10 wt% hBN/PAN nanofiber delivers the highest discharge capacity of 144 mAh g^{-1}. The high porosity and ionic conductivity properties of the hBN/PAN composite nanofiber are considered as the key factors for obtaining improved cell capacities.

Cycling performances of the cells assembled with different nanofiber separators are demonstrated in Fig. 8. From Fig. 8 it is observed that among all assembled cells the cell containing PAN nanofiber separator shows the lowest capacity retention, which is 85.8% after 50 cycles. On the other hand, the capacity retentions of the cells containing 7 wt% hBN/PAN and 10 wt% hBN/PAN composite nanofiber are 88.3%, and 92.4%, respectively. The improved cycling performance of the cells containing hBN/PAN composite nanofiber materials can be attributed to unique porous structure, high liquid electrolyte uptake, and low interfacial resistance of these composite nanofiber electrolytes. Similar cycling performance improvement was also reported by Yanilmaz et al. [39], who compared the cycling performance of the SiO_2 nanoparticle mixed nylon 6,6 composite nanofiber separator with a pure nylon 6,6 nanofiber separator.

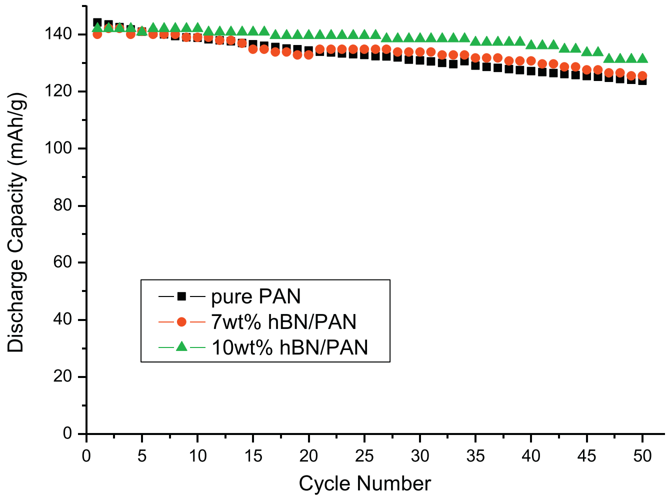

Fig. 8. Cycling performance of Li/GPE/$LiCoO_2$ cells containing electrolyte loaded hBN/PAN nanofiber with different hBN contents at 0.1C.

4. Conclusion

Novel hBN/PAN electrospun fibers were produced by using electrospinning method to apply as thermally stable Li-ion battery separator and the host of a gel polymer electrolyte. SEM analysis verified the presence of greatly porous nanofibrous structure. Experimental results indicated that the electrospun 10 wt% hBN/PAN composite fibers has better thermal stability (280 °C), largest electrolyte uptake (1250%), highest ionic conductivity (1.0×10^{-3} Scm^{-1}), and best electrochemical stability (4,7 V). The cells containing hBN/PAN nanofiber separators with high hBN contents showed excellent cycling and C-rate performance. Thus it can be proposed that hBN/PAN nanofibers are promising separator and the host of a gel polymer electrolyte candidate for high-performance Li-ion batteries.

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